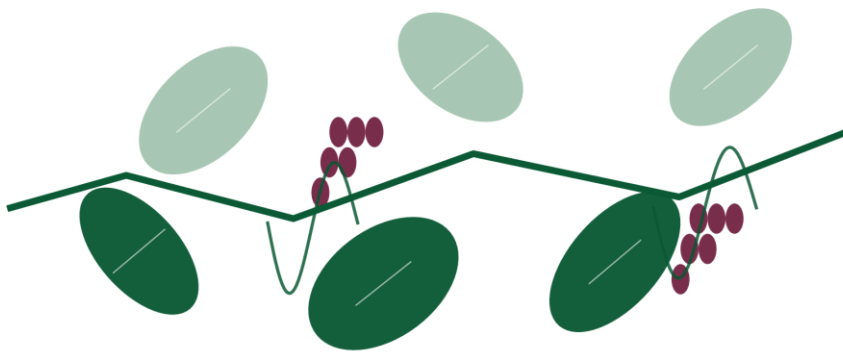


Predicting Annual Wine Production by Viticultural Region in Portugal

Final Project Report

Master's in Green Data Science 2025-2026



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UC | **Practical Machine Learning**

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Abstract. This project develops and evaluates a reproducible machine-learning workflow for forecasting annual wine production across Portuguese viticultural regions. The work uses official Instituto da Vinha e do Vinho (IVV) production and vineyard-area datasets included in the repository. The modelling task is treated as a temporal regression problem: rolling-origin validation over 2018-2022 is used for model selection, while 2023-2025 is held out as an untouched chronological test period. A second, experimental one-step-ahead workflow introduces lagged production features and supports the deployed Hugging Face Gradio application. The final interpretation is deliberately cautious: simple baselines remain essential, lagged information improves the experimental one-year-ahead forecast, and the deployed prediction bands should be read as empirical uncertainty indicators rather than formal confidence intervals.

1. Introduction and problem statement

Wine production is an economically and agronomically relevant activity in Portugal, but regional production varies strongly across campaigns. This variability creates a practical forecasting problem for planning, stock management, and communication of expected production. The objective of this project is to estimate annual total wine production, measured in hectolitres, for each Portuguese viticultural region using tabular historical data.

The project was designed around two technical questions. The first asks how far a restricted, leakage-resistant model can go using only region, campaign year, and vineyard area. The second asks whether standard regression models improve on transparent temporal baselines. This second question is important because high test R^2 can be caused by persistent scale differences between regions rather than by accurate prediction of year-to-year variation. For this reason, the repository compares machine-learning models against persistence and historical-mean baselines instead of evaluating a model in isolation.

2. Data sources, cleaning, and feature engineering

The repository contains raw and processed IVV datasets. The processed wine-production table contains 238 region-year observations for 14 viticultural regions from 2009 to 2025. The processed vineyard-area table contains 288 region-year observations from 1989 to 2025. The target variable is `total_production_hl`. The strict evaluation workflow uses `region`, `year_start`, and `vineyard_area_ha` as predictors; the experimental future-forecast workflow additionally constructs lagged production variables from previous campaigns.

Cleaning was implemented in the repository notebooks. The raw workbooks required standardisation of region labels and campaign labels, conversion of Portuguese-formatted numbers, reshaping of production subtables into one row per region and campaign year, checks for unique region-year keys, and exclusion of implausible values. Production and vineyard-area data were joined by region and campaign start year. Production components such as DOP, IGP, varietal/year production, and non-certified production were not used as predictors in the final modelling pipeline because they belong to the same target campaign and would leak information about the total production being predicted.

The relevant data types are categorical region labels, integer campaign years, continuous vineyard area in hectares, and continuous wine production in hectolitres. Missing vineyard-area values are handled inside scikit-learn pipelines using training-set median imputation with a missingness indicator. This keeps

preprocessing inside the temporal validation loop and avoids using validation or test information when estimating imputation parameters.

Table 1. Processed datasets used in the final repository workflow.

Dataset	Processed rows	Temporal coverage	Main fields	Use in modelling
Wine production	238	2009-2025	region; year_start; total_production_hl; production components	Target variable and lag construction
Vineyard area	288	1989-2025	region; year_start; vineyard_area_ha	Predictor and scenario input in the app

3. Data organization and validation strategy

The modelling problem is temporal, so random splitting was avoided. In a random split, future campaigns could indirectly influence models evaluated on earlier campaigns, and observations from the same year could be divided across training and test sets. The project therefore uses chronological evaluation. During development, expanding rolling-origin validation predicts each validation year from all preceding years. After selecting the model on validation MAE, the final assessment is performed on the untouched years 2023, 2024, and 2025.

This design follows the intended forecasting use case: at prediction time, only past campaigns are available. It also accounts for the fact that observations are not independent in the usual random-sample sense; each region appears repeatedly across time, so within-region history is informative and must be handled consistently.

Chronological evaluation: no random splitting and no future information in training folds

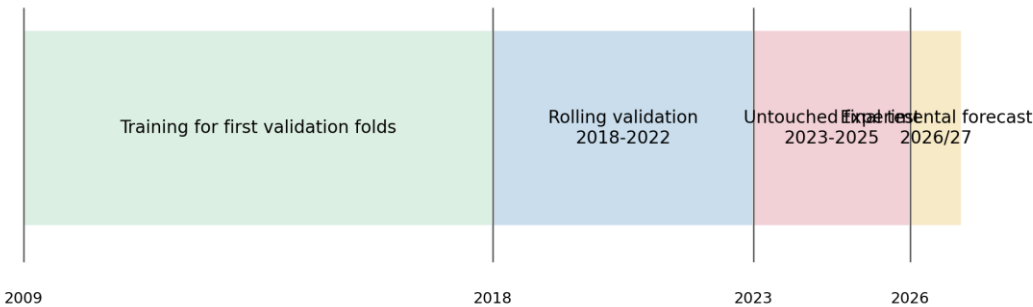


Figure 1. Chronological validation design used in the project. Rolling-origin validation is used for model selection, and 2023-2025 remains an untouched final test period.

4. Methods and modelling pipeline

Two transparent baselines are implemented. The persistence baseline predicts the most recent observed production for the same region; the historical regional mean baseline predicts the mean of all previous production observations for that region. These baselines are not auxiliary: they define the minimum performance that a practical machine-learning model should improve upon.

The machine-learning candidates are ordinary least-squares linear regression, Ridge regression with alpha values 0.1, 1, 10, and 100, and random-forest regression with 500 trees, max_features=0.8, a fixed random seed, and minimum leaf sizes 1, 2, and 4. The scikit-learn pipeline applies median imputation and scaling to numerical predictors and one-hot encodes the region variable. Model selection uses MAE in the rolling-

validation period because it is directly interpretable in hectolitres. RMSE is reported to expose large errors, and R2 is reported to describe explained variation on each evaluation split.

The experimental future-forecast workflow uses a different information set. It adds `production_lag_1`, `production_lag_2`, `production_mean_3`, and `production_trend_1`, all shifted within each region before the target year. This design enables a one-step-ahead 2026/27 simulator, but it must be interpreted separately from the stricter fixed-origin evaluation because it assumes that the immediately preceding campaign is known.

Table 2. Modelling workflows implemented in the repository.

Workflow	Predictors	Evaluation question	Main output
Strict fixed-origin evaluation	region; year_start; vineyard_area_ha	Can basic tabular predictors improve on simple baselines for future years?	model_comparison.csv and test_predictions.csv
Experimental lagged workflow	region; year_start; vineyard_area_ha; production lags and rolling mean/trend	Can recent regional production history improve a one-step-ahead forecast?	future_model_comparison.csv and forecast_2026_27.csv

5. Results

5.1 Strict fixed-origin evaluation

In the strict evaluation, linear regression achieved the lowest rolling-validation MAE among the candidate models, with validation MAE of 58,180 hl. However, final performance on the untouched 2023-2025 test period was better for the persistence baseline, which obtained test MAE of 55,139 hl. The selected linear model obtained 78,982 hl MAE. This result is central to the project because it shows that a model can perform well by R2 while still failing to beat a very simple operational baseline in absolute error.

Table 3. Main strict evaluation results. Validation uses rolling-origin 2018-2022; test uses 2023-2025.

Model	Validation MAE (hl)	Test MAE (hl)	Test RMSE (hl)	Test R2
Persistence - last region value	70,220	55,139	98,771	0.961
Historical region mean	58,464	70,377	117,842	0.945
Linear regression	58,180	78,982	115,420	0.947
Ridge (alpha=0.1)	59,695	79,058	116,690	0.946
Random forest (min_leaf=1)	66,512	102,482	237,583	0.777

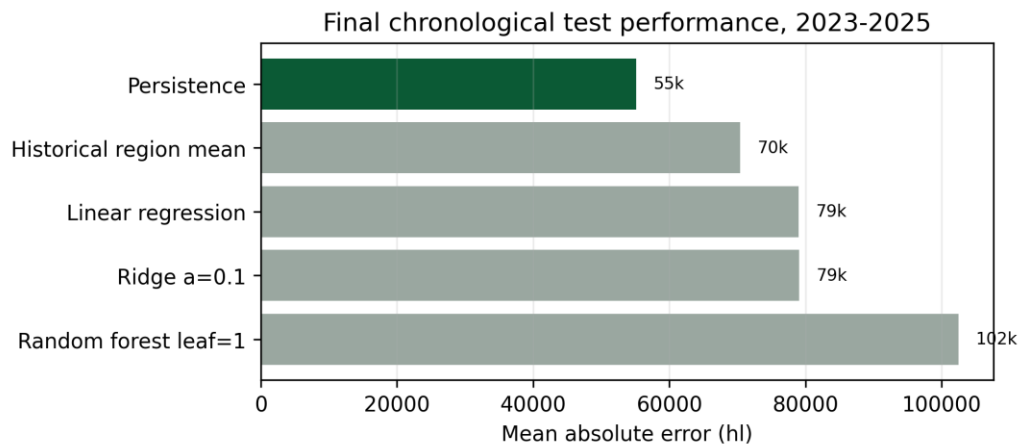


Figure 2. Final test MAE for the strict fixed-origin workflow. The persistence baseline has the lowest MAE on 2023-2025.

5.2 Experimental one-step-ahead lagged workflow

The lagged workflow produces a more operational one-year-ahead simulator. Here, the random forest with lag features was selected by validation MAE and also performed best in the 2023-2025 one-step-ahead test. Its test MAE was 59,026 hl, compared with 69,403 hl for previous-campaign persistence. This improvement is meaningful only within the one-step-ahead information set, because the model can use the latest available campaign and lagged regional history.

Table 4. One-step-ahead lagged workflow results used by the experimental forecast simulator.

Model	Validation MAE (hl)	Test MAE (hl)	Test RMSE (hl)	Test R2
Random forest with lags	52,314	59,026	100,230	0.960
Persistence - previous campaign	70,220	69,403	126,723	0.936
Ridge with lags (alpha=10)	56,308	78,821	124,083	0.939
Ridge with lags (alpha=1)	56,040	81,914	128,748	0.934
Linear regression with lags	62,814	88,248	120,600	0.942

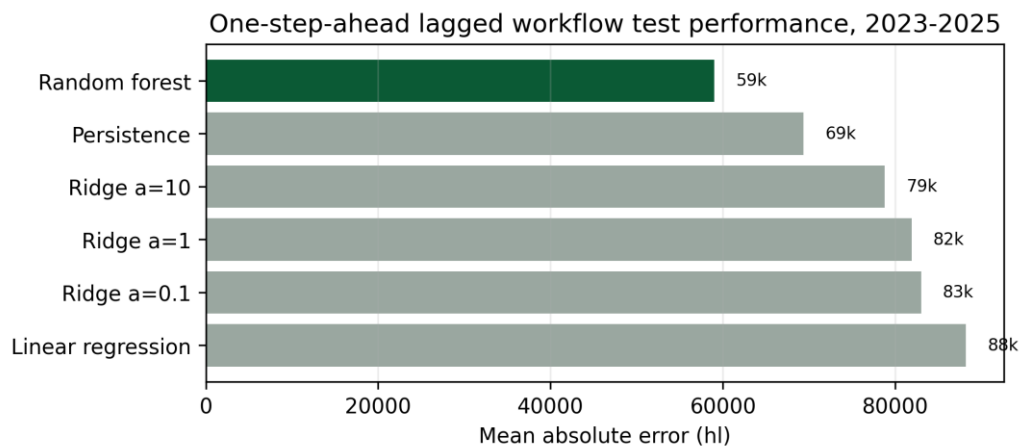


Figure 3. Final one-step-ahead test MAE for the lagged workflow. The selected random forest with lag features improves on previous-campaign persistence in this setting.

5.3 Experimental 2026/27 forecast output

After evaluation, the selected lagged model was refitted through 2025 and used to produce the experimental 2026/27 forecast table. The default scenario uses the latest vineyard-area values present in the data. The lower and upper values are empirical uncertainty bands derived from validation errors and practical lower-bound safeguards; they are not formal probabilistic confidence intervals.

Table 5. Selected 2026/27 forecast rows from forecast_2026_27.csv. The complete table is available in the repository outputs.

Region	Vineyard area (ha)	Prediction (hl)	Lower band (hl)	Upper band (hl)
Douro	43,929.72	1,411,028	1,263,052	1,559,003
Lisboa	17,819.98	1,166,580	1,018,605	1,314,556
Alentejo	25,421.93	1,068,766	920,791	1,216,742
Verdes	23,272.15	1,054,430	906,455	1,202,406
Tejo	11,713.12	637,021	489,045	784,996
P. Setúbal	7,355.33	477,691	329,716	625,667
Bairrada	9,180.68	188,368	40,393	336,344
Beira Interior	11,360.82	219,067	71,091	367,043

6. Technical analysis and interpretation

The strict evaluation shows the value of baselines. The high R^2 values in several models mainly reflect stable structural differences between regions, not necessarily precise forecasting of annual movements. For example, large-producing regions are consistently separated from smaller regions, which improves R^2 even when year-to-year deviations are not fully captured. The persistence baseline performs strongly because recent production contains short-term regional information that a model using only region, year, and vineyard area cannot recover.

The lagged workflow addresses part of this limitation by explicitly adding recent production history. The improvement of the random forest with lags suggests that short-term state variables matter more than a common linear time trend. Nevertheless, the number of observations remains small for a panel forecasting problem, and the predictor set does not include weather, disease pressure, harvest timing, irrigation, management practices, or market-related factors. The project should therefore be read as a reproducible forecasting prototype rather than as a complete operational forecasting system.

The prediction bands used in the simulator are empirical. They summarize errors observed during validation and are useful for communicating uncertainty, but they should not be described as confidence intervals. In particular, lower-bound safeguards in the app prevent negative or implausibly low lower values; these safeguards improve user-facing plausibility but are practical interface rules rather than statistical guarantees.

7. Deployment and reproducibility

The practical workflow is organized in the GitHub repository. The main reproducible scripts are `src/evaluate_models.py` for the strict temporal model comparison, `src/future_forecast.py` for the experimental lagged one-step-ahead forecast, and `src/app.py` for the Gradio interface. The repository also contains raw and processed data, notebooks for cleaning and exploratory analysis, generated output tables, generated figures, a requirements file, and a README with commands for reproducing the evaluation.

The Hugging Face Space wine-production-predictor is an optional deployment of the experimental workflow. It loads the processed data, evaluates the forecasters, displays validation and test metrics, allows the user to select a region, and allows vineyard-area scenario changes. The app demonstrates the practical pipeline and makes the 2026/27 forecast explorable, but it is not an independent validation source and should not be interpreted as a formal decision-support tool.

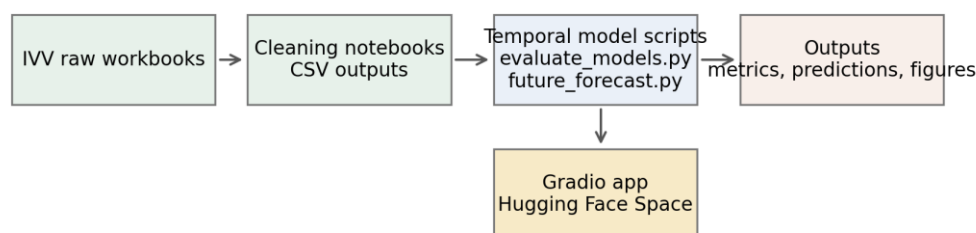


Figure 4. Practical organization of the repository and Hugging Face deployment. The app is a lightweight interface built on the same processed data and forecast functions.

8. Limitations and conclusions

The main limitations are the small number of region-year observations, the aggregated regional scale, and the limited predictor set. Vineyard area changes slowly and helps explain regional scale, but it cannot explain annual shocks. The lagged workflow improves one-step-ahead performance, but this improvement depends on using previous production values; it does not prove a causal relationship between vineyard-area scenarios and production. Scenario changes in the app should therefore be treated as exploratory assumptions.

The project delivers a leakage-resistant comparison between baselines and machine-learning regressors, a second lagged forecasting workflow, reproducible outputs, and an optional Gradio deployment. The strongest technical conclusion is not that machine learning automatically outperforms simple methods. Rather, the work shows that evaluation design and information sets are decisive: under the strict feature set, persistence is the best final-test method; under the lagged one-step-ahead setting, the random forest with recent production history performs best.

9. References

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10. Contributions

All members contributed to the conceptual discussion, interpretation of results, and written preparation of the project.

Andrea Dombe: IVV source review and raw-data organization; support in checking the consistency of production and vineyard-area records; exploratory analysis of regional patterns; review of output tables and contribution to the written project materials.

Dandara França: Support in preparing model-ready datasets after the cleaning stage; exploratory analysis and review of visual outputs; discussion of baseline and machine-learning model comparisons; interpretation of model results; implementation and revision of `src/app.py` for the Hugging Face/Gradio deployment; and contribution to written project materials.

Fernanda Chácara: Data cleaning and preprocessing workflow; development of the forecast simulator notebook; implementation and technical revision of `src/future_forecast.py`, including lagged features, temporal validation logic and future forecast workflow; final technical review of the report and preparation of presentation materials.